OIL CONTROL VALVE (OCV) [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (WITH 4-1 EXHAUST SYSTEM)]

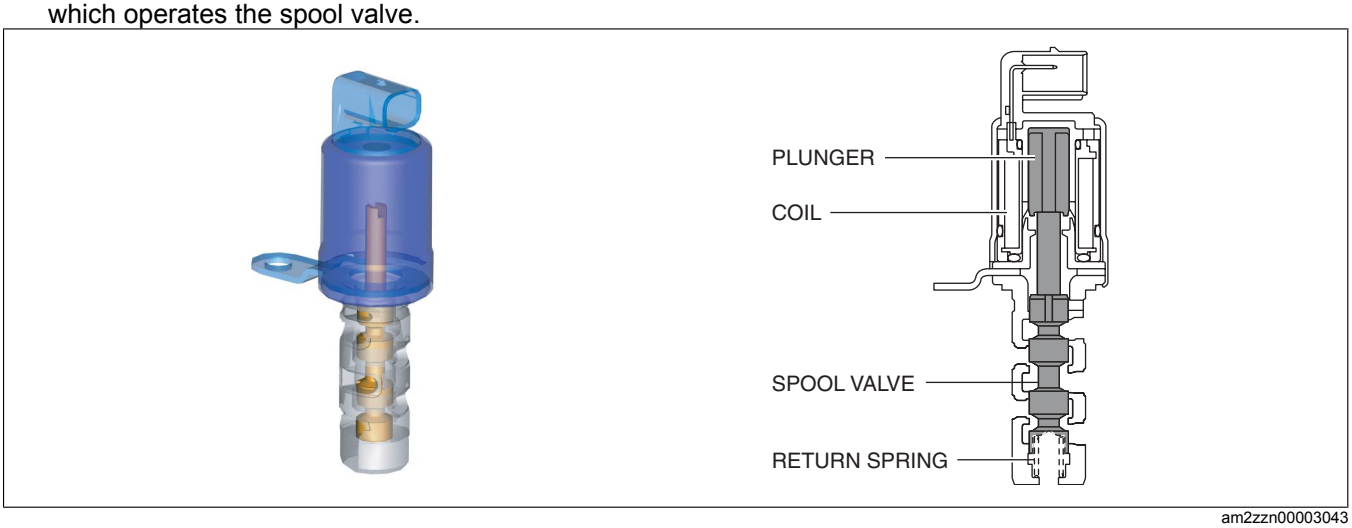
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Purpose, Function

- The OCV operates according to a signal from the PCM depending on the engine operation conditions and switches the hydraulic passage to the hydraulic variable valve timing actuator.

Construction

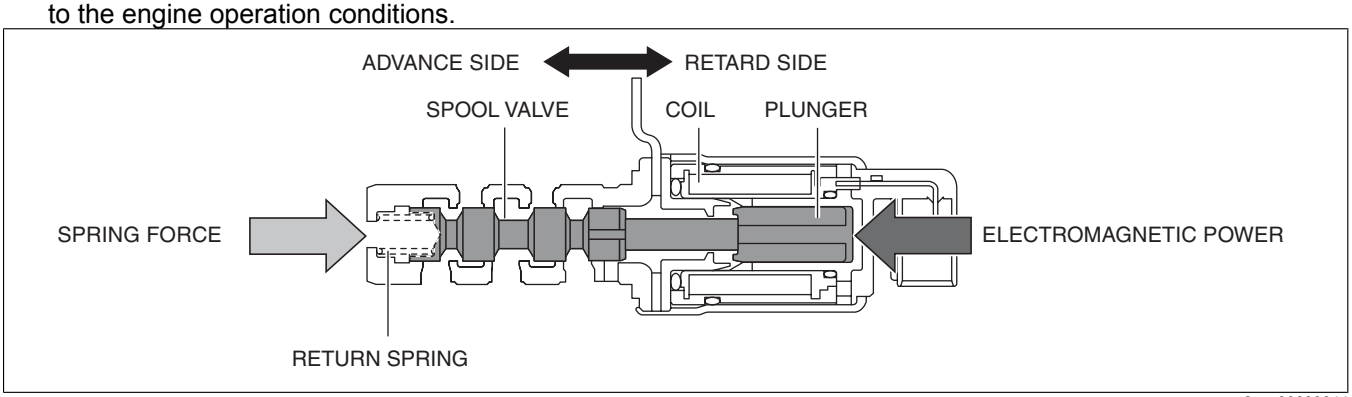
- The OCV is positioned on the front camshaft cap.
- The OCV consists of a spool valve which switches the engine oil passage and a plunger, coil, and a return spring which operates the spool valve.



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Operation

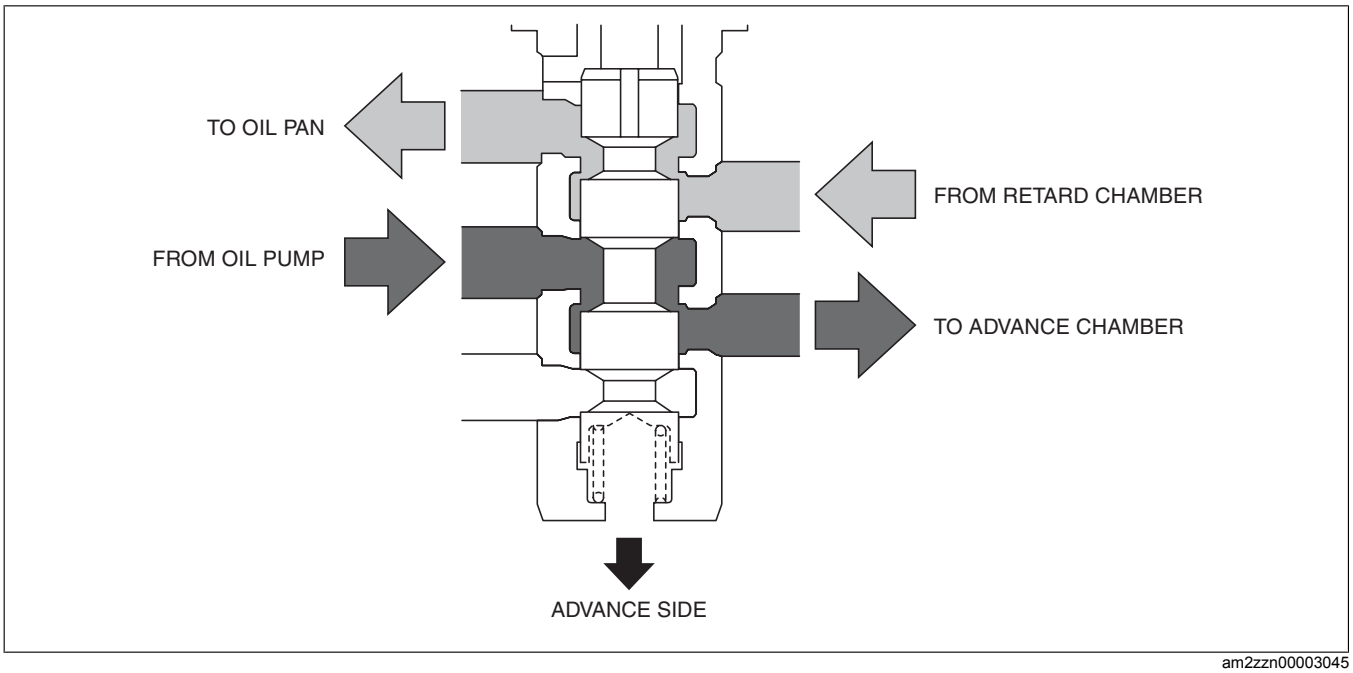
- The spool valve moves to the retard side by the spring force of the return spring during non-energization of the OCV. When the coil is energized by a signal from the PCM, electromagnetic power, which moves the plunger to the advance side, is generated and the spool valve moves to the advance side. The spool valve position balances the electromagnetic power with the spring force.
- The PCM performs duty-control of a signal to the coil so that the spool valve is in the optimum position according to the engine operation conditions.



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Advance mode

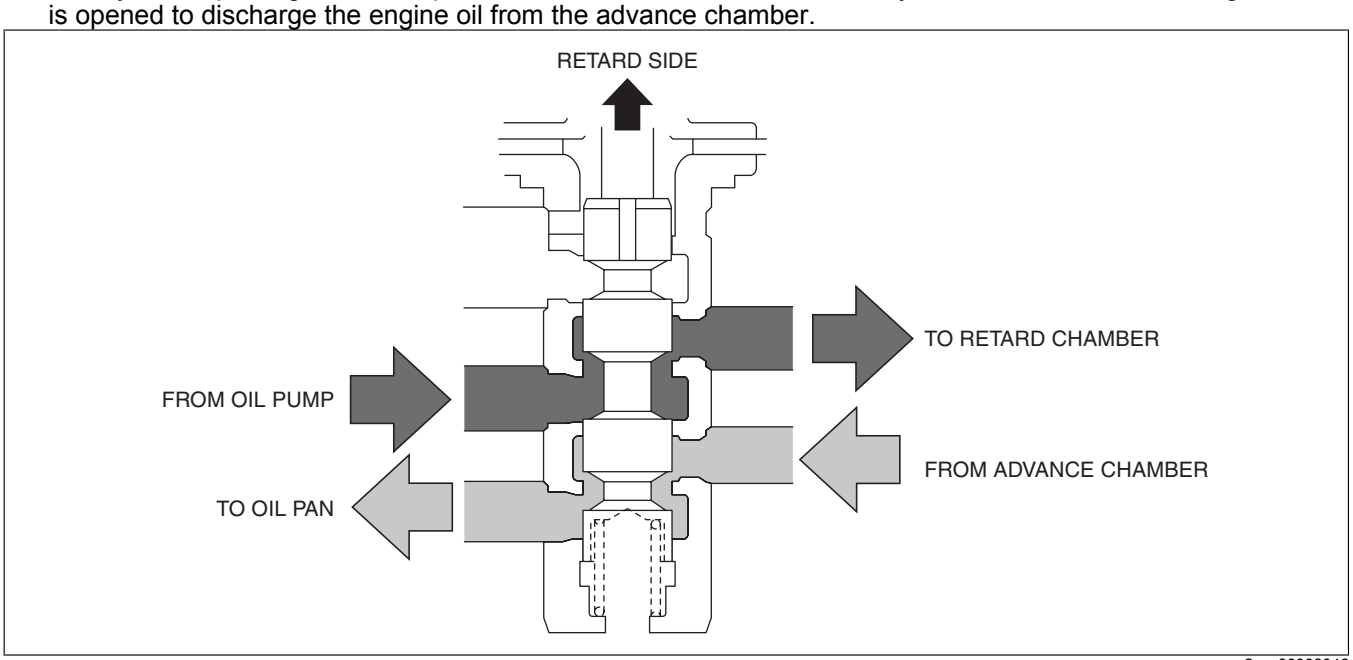
- The spool valve moves to the advance side by a signal from the PCM.
- The hydraulic passage to the advance chamber of the hydraulic variable valve timing actuator from the oil pump is opened to supply engine oil to the advance chamber.
- The hydraulic passage to the retard chamber of the hydraulic variable valve timing actuator from the oil pump is closed.
- The hydraulic passage to the oil pan from the retard chamber of the hydraulic variable valve timing actuator is opened to discharge the engine oil from the retard chamber.



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Retard mode

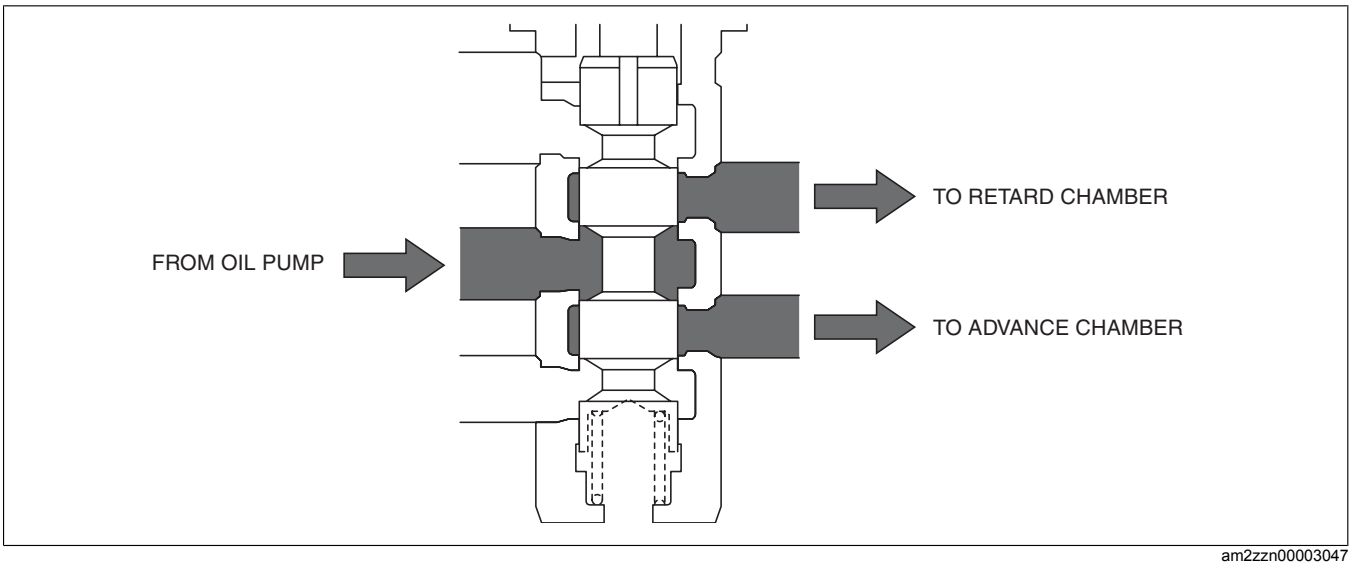
- The spool valve moves to the retard side by a signal from the PCM.
- The hydraulic passage to the retard chamber of the hydraulic variable valve timing actuator from the oil pump is opened to supply engine oil to the retard chamber.
- The hydraulic passage to the advance chamber of the hydraulic variable valve timing actuator from the oil pump is closed.
- The hydraulic passage to the oil pan from the advance chamber of the hydraulic variable valve timing actuator is opened to discharge the engine oil from the advance chamber.



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Hold mode

- The spool valve is positioned in the area between the advance and retard by a signal from the PCM. (Middle point maintenance)
- For the middle point maintenance, the hydraulic passages to the advance and retard chambers of the hydraulic variable valve timing actuator from the oil pump is opened to supply a small amount of engine oil to the advance and retard chambers.



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Fail-safe

DTC	Fail-safe
P2088:00	• Set the intake variable valve timing control to the maximum retard position.
P2089:00	